

CSC1202 Project — LLM-Based Agents

The project will explore how LLM-based agents can be used to improve different aspects of life at DCU. Each group will identify a meaningful problem, need, or activity within the university context and develop an LLM-based agent to address it. Projects may relate to areas such as learning and teaching, research, student experience and support, or university services and activities. Students are also welcome to propose their own project ideas, provided that they fit the scope and objectives of the module.

Through the project, students will gain practical experience in the design, implementation, and evaluation of LLM-based agents. Depending on the selected problem, the project may involve capabilities and techniques such as reasoning and planning, tool use and function calling, memory and context, workflow coordination, human interaction and control, and multi-agent collaboration.

Students will consider the design and implementation of the agent, its interaction with tools and other components, how its capabilities and behaviour are evaluated, and relevant issues such as reliability, limitations, risks, and safety.

The project will be organised around four assessed components. Page limits apply to the main reports. Where necessary, additional supporting material may be included in appendices. Appendices should contain supporting details and should not be used to replace material that belongs in the main report.

1. Project Scoping — 10%

Provide a brief summary of the proposed project in your own words. The summary should describe the problem or opportunity being addressed, the proposed LLM-based agent, and the main project goals. The purpose of this component is to define the high-level scope and goals of the project.

The report should be a maximum of 1 page and is due on Friday of Week 4.

Detailed submission guidelines and the marking scheme will be provided separately.

2. Project Specification and Initial Design — 20%

Building on the project scoping, describe the proposed agentic system in more concrete terms. Specify the intended users and use case, the main tasks the agent will perform, the tools or external resources it may use, the expected level of autonomy and human involvement, and important initial risks or constraints.

Provide an initial architecture diagram showing the main components of the proposed system and how they interact, including the LLM/agent, tools, external resources, memory or other components where applicable. Include an initial plan for developing the project.

The report should be a maximum of 4 pages and is due on Friday of Week 7.

Detailed submission guidelines and the marking scheme will be provided separately.

3. Implementation and Evaluation — Report, Presentation and Demo — 60%

Implement the proposed LLM-based agent and evaluate the completed system. The report should describe the implementation of the main agent capabilities, interactions with tools and other components, and any significant changes from the initial specification and architecture.

Students should systematically evaluate the agent using appropriate test scenarios, criteria and metrics, analyse its behaviour and limitations, and consider relevant reliability, risk and safety issues.

The deliverables are a working prototype, an implementation and evaluation report of a maximum of 7 pages, and a final presentation. The presentation will include a demonstration of the working system. This component is due on Friday of Week 11.

Detailed submission guidelines and the marking scheme will be provided separately.

4. Project Meeting Reports — 10 marks

Each group is required to submit a short report (max 400 words) for **four assessed project meetings**, due on Tuesday of Weeks **4, 6, 8, and 10**. Groups are encouraged to **meet more frequently as needed** throughout the project; however, only these four meeting reports will be assessed.

Each report should briefly record:

- project progress;
- work discussed and agreed during the meeting;
- planned next steps; and
- group members present or absent.

Each meeting report is worth **2.5 marks**, giving a total of **10 marks** across the four assessed meetings.

Marking Scheme — Each Meeting Report [2.5 marks]

Meets expectations [2.5 marks]	Partially meets expectations [1.5 marks]	Does not meet expectations [0 marks]
The report clearly records project progress, work discussed and agreed, planned next steps, and attendance.	The report records the meeting and project progress, but one or more required elements are missing, unclear, or insufficiently described.	The report is not submitted, or provides no meaningful record of the meeting or project progress.

Group Formation and Participation

The project will be completed in groups of 5–6 students. Groups must be formed by Friday of Week 2. Each group should provide the names, student numbers, and DCU email addresses of all group members.

How to join your group in Loop:

- 1) Go to the Loop module/course page.
- 2) Open the activity called “Choose Your Group”
- 3) You will see a list of the available groups (Group Numbers).
- 4) **Choose ONE group** that you would like to join. Discuss among your intended group members beforehand.
- 5) Click on **Save my Choice** to confirm your selection.
- 6) Your name will then be added to the group.

Important:

- Please choose only one group.
- Check that you have selected the group you actually want before submitting.
- If a group is full, you will not be able to select it.
- Students who have not joined a group by the deadline will be automatically allocated to a group.

All group members are expected to contribute actively and consistently to the project. Each project deliverable must include a contribution table identifying the contribution of each group member to that deliverable. The project meeting reports will also record attendance and provide evidence of the allocation and progress of project work.

A student who does not contribute to a particular project component may receive zero for that component. Being listed as a member of a group does not automatically entitle a student to the group mark.

Any significant problems concerning participation or contribution should be raised as early as possible and should not be left until the final submission.

Submission Information

The information above provides the general guidelines and expected deliverables for the project. More detailed instructions will be provided in due course for group formation, each project deliverable, and the project meeting reports. These will include the specific submission requirements, required format and content, submission process, and marking scheme for each assessed component.